Distance Distortion in Binary Bivariate Bicycle Double Covers

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Abstract

Binary bivariate bicycle codes with three distinct monomials in each check need not obey either of the proposed double-cover distance inequalities $d \le d_2 \le 2d$. We give connected, term-matched covers with exact parameters $[[78, 4, 6]] \to [[156, 4, 14]]$ and $[[156, 4, 14]] \to [[312, 8, 12]]$. Explicit logical operators and two independent exhaustive circuit enumerations certify both examples. We then prove the replacement inequalities $d \le 4d_2^3$ and $d_2 \le 8d^3$, for every positive-dimensional strict-trinomial binary abelian double cover. The upper constant follows from an index-two transport of incidence relation lattices, combining the two previously separate proof directions. The proof treats arbitrary finite-support circuits, repeated roots, torsion collapse, and finite homology not represented by surviving compact classes. A self-contained cleaning and geometry-of-numbers argument also gives an explicit $O(n^{3/4})$ ambient distance estimate and $O(\sqrt{N})$ truncated distance-distortion bounds. Projection and transfer identify the mechanisms behind the counterexamples; in particular, $k_2 = 2k$ still implies $d_2 \le 2d$. All claimed finite distances are accompanied by regenerable, independently checked certificates. Neither a universal linear amplification bound nor a positive universal degradation ratio is asserted.

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1 Two counterexamples to the double-cover distance bounds

Symons, Rajput, and Browne define term-matched group-algebra covers of bivariate bicycle (BB) codes and conjecture $d \le d_h \le hd$ for an h-cover [1, Theorem 3.1 and Section 4.3]. Their proved odd-degree statements do not cover binary double covers. We begin with two counterexamples to the conjectured even-degree extension. The code appearing upstairs in the first example is exactly the code appearing downstairs in the second.

1.1 Conventions and the three codes

For a finite abelian group G, put $R = \mathbb{F}_2[G]$ and use its group elements as the physical coordinate basis. A pair of commuting polynomials gives the complex

$$R \xrightarrow{\partial_2} R^2 \xrightarrow{\partial_1} R, \qquad \partial_2 h = (Bh, Ah), \qquad \partial_1(u, v) = Au + Bv. \tag{1}$$

If M_P is the multiplication matrix of P, our CSS matrices are

$$H_X = (M_A \; M_B), \qquad H_Z = (M_B^T \; M_A^T). \tag{2}$$

Thus $H_X H_Z^T = 0$. A Z logical is a cycle not in $\operatorname{im} \partial_2$. We always minimize Hamming weight in the displayed group basis, not in an abstract module basis. The inversion-and-block-swap permutation

$$J(u, v) = (v^\star, u^\star), \qquad g^\star = g^{-1}, \tag{3}$$

identifies the Z and X homologies isometrically. Consequently $d_X = d_Z = d$.

Write $Q(A, B; \ell, m)$ when $G = \mathbb{Z}_\ell \times \mathbb{Z}_m$. The monomial $x^i y^j$ has index $im + j$ in each block; the right block has offset ℓm. Every polynomial below has exactly three distinct supported monomials. Period $m = 1$ means $y = 1$, not a lost monomial: the three x powers remain distinct. This case is explicitly allowed by the group-algebra definition and occurs among the examples in [1, Remark 5 and Tables 2–4].

Set

$$A = 1 + x^8 + x^{25}, \qquad B = 1 + x^2 + x^{13}, \tag{4}$$

and, for the second cover, set

$$A' = 1 + x^8 y + x^{25}, \qquad B' = 1 + x^2 y + x^{13}. \tag{5}$$

Theorem 1.1 (Two exact, computer-assisted counterexamples)**.** *The following are connected, valid term-matched binary BB double covers:*

$$Q(A, B; 39, 1) = [[78, 4, 6]] \longrightarrow Q(A, B; 78, 1) = [[156, 4, 14]], \tag{6}$$
$$Q(A, B; 78, 1) = [[156, 4, 14]] \longrightarrow Q(A', B'; 78, 2) = [[312, 8, 12]]. \tag{7}$$

They give $d_2/d = 7/3 > 2$ *and* $d_2/d = 6/7 < 1$*, respectively.*

The algebraic parts of the proof follow below. The only computer-assisted parts are the complete short-circuit exclusions, whose branching completeness and pruning safety are proved in Appendix A. Two standalone implementations regenerate, rather than read, each complete circuit set.

For (6), the covering map is $x^i \mapsto x^{i \bmod 39}$ and the polynomials are unchanged. For (7), it is $x^i y^j \mapsto x^i$. Terms match individually in both cases. The induced maps on the four vertex types of the full Tanner graph have two preimages per vertex and are bijective on each neighborhood. In particular these are graph covers, not merely coefficient reductions that happen to commute with a differential. Constant terms connect all vertex types within a cell. Since $\gcd(8, 25, 39) = \gcd(8, 25, 78) = 1$, the first two graphs are connected. In the last graph, x^{25} and x^{13} generate all x shifts, and $x^8 y$ supplies the remaining y shift.

1.2 Dimensions

Let $g = 1 + x + x^2$ and $F_L = 1 + x^L$. Polynomial division gives $\gcd(A, B) = g$. For an explicitly checkable Bezout identity $UA + VB = g$, the supports are

$$\operatorname{supp} U = \{1, 2, 4, 5, 6, 9, 10\},$$
$$\operatorname{supp} V = \{0, 2, 5, 6, 7, 8, 9, 10, 11, 13, 14, 16, 17, 18, 21, 22\}.$$

Since $g \mid F_{39}, F_{78}$, the cokernel of ∂_1 has dimension two in both cyclic codes. In general, inversion identifies the two check ranks and gives

$$k = 2 \dim_{\mathbb{F}_2} R/(A, B). \tag{8}$$

Thus both cyclic codes have $k = 4$. This argument includes the repeated-root period 78.

For the $(78, 2)$ code, eliminate y using $B' = 0$:

$$y = x^{-2}(1 + x^{13}), \qquad C = A + x^6 B = (1 + x^6)(1 + x^{19}).$$

It follows that

$$\mathbb{F}_2[x, y]/(F_{78}, 1 + y^2, A', B') \cong \mathbb{F}_2[x]/(F_{78}, B^2, C).$$

The common gcd is $g^2 = 1 + x^2 + x^4$, so the quotient has dimension four and $k_2 = 8$. Independent binary elimination gives the same ranks:

Code	Qubits	$\operatorname{rank} H_X$	$\operatorname{rank} H_Z$	k
$Q(A, B; 39, 1)$	78	37	37	4
$Q(A, B; 78, 1)$	156	76	76	4
$Q(A', B'; 78, 2)$	312	152	152	8

No minimum-distance conclusion is inferred from these ranks.

1.3 A weight-six base logical and a weight-fourteen cover logical

Put $F = 1 + x^{39}$ and

$$z = (z_L, z_R) = (1 + x^8 + x^{16} + x^{25},\ 1 + x^{11}), \qquad c = x^2 + x^{11}.$$

Ordinary polynomial multiplication gives

$$Az_L + Bz_R = Fc. \tag{9}$$

The two coordinates of z are both x modulo g, whereas every boundary coordinate is zero modulo g. Thus z is a weight-six nonboundary modulo F.

Now let $P = x^9 + x^{22} + x^{35} + x^{37}$. The further identity

$$Az_L + BP + c = F(1 + x^2 + x^9)$$

shows that

$$w_{14} = (x^{39} z_L,\ z_R + FP) \tag{10}$$

is a cycle modulo $F^2 = 1 + x^{78}$. Its supports are

$$\begin{aligned} \operatorname{supp}(w_{14})_L &= \{39, 47, 55, 64\}, \\ \operatorname{supp}(w_{14})_R &= \{0, 9, 11, 22, 35, 37, 48, 61, 74, 76\}. \end{aligned}$$

Its projection is the literal chain z. Since projection takes boundaries to boundaries, w_{14} is a nonboundary. This proves the upper estimates $d \le 6$ and $d_2 \le 14$ for (6).

The tempting weight-twelve transfer Fz is instead a boundary. To make the obstruction concrete, let H have support

$$\begin{aligned} \{&1, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 23, 24, 25, 27, 28, 31, 32, \\ &34, 37, 38, 41, 43, 44, 46, 48, 51, 53, 54, 57, 61, 63, 65, 66, 68, 74\}. \end{aligned}$$

Then

$$(BH, AH) = Fz \pmod{F^2}. \tag{11}$$

This is an actual two-chain filling, not an argument based only on a vanishing homological label.

1.4 A weight-twelve logical in the distance-decreasing cover

In $\mathbb{F}_2[x, y]/(1 + x^{78}, 1 + y^2)$ define

$$\sigma = 1 + x^{39} y, \qquad \zeta_L = 1 + x^8 y + x^{16} + x^{25}, \qquad \zeta_R = 1 + x^{11} y,$$

and set $w_{12} = \sigma(\zeta_L, \zeta_R)$. Before reduction of the x period,

$$A'\zeta_L + B'\zeta_R = \sigma y(x^2 + x^{11}), \qquad \sigma^2 = 1 + x^{78}.$$

Hence w_{12} is a cycle. Its two supports, in exponent-pair coordinates, are

$$\begin{aligned} L &: \{(0,0), (8,1), (16,0), (25,0), (39,1), (47,0), (55,1), (64,1)\}, \\ R &: \{(0,0), (11,1), (39,1), (50,0)\}. \end{aligned}$$

All twelve positions are distinct.

For a direct nonboundary certificate use

$$T = \mathbb{F}_4[\eta]/(\eta^2), \quad \omega^2 + \omega + 1 = 0, \qquad \phi(x) = \omega + \eta, \quad \phi(y) = 1 + \omega\eta.$$

These images satisfy the two period relations and

$$\phi(A') = \phi(B') = 0, \quad \phi(\sigma) = \eta, \quad \phi(\zeta_L) = \omega, \quad \phi(\zeta_R) = \omega \pmod{\eta}.$$

Both boundary coordinates map to zero, but

$$\phi((w_{12})_L) = \phi((w_{12})_R) = \omega\eta \neq 0. \tag{12}$$

Thus w_{12} is nontrivial. Notice that its projection is exactly $Fz = (BH, AH)$ from (11). The short cover class therefore projects to a base boundary, as it must in a distance-decreasing example.

1.5 Complete distance exclusions

A minimum-weight nonboundary has circuit support: a proper subcycle is either a smaller nonboundary or a boundary whose removal shortens the same nonzero class. Every column of either check has odd weight three, so every cycle has even weight. To certify the three distances it is therefore enough to exclude nonboundary circuits through weights $4, 12, 10$, respectively.

The independently regenerated complete Z-circuit counts are

Code	Search cutoff	Weight 6	Weight 10	Weight 12	Total
$Q(A, B; 39, 1)$	4	—	—	—	0
$Q(A, B; 78, 1)$	12	78	468	195	741
$Q(A', B'; 78, 2)$	10	156	936	—	1092

Every enumerated circuit is a boundary. For the middle code, their translation orbits are precisely the orbits of

$$\partial_2(1), \quad \partial_2(1 + x^a) \ (a \in \{2, 8, 11, 13, 17, 25\}), \quad \partial_2(A), \quad \partial_2(B), \quad \partial_2(H).$$

The first nine orbits have size 78, while $\partial_2(H) = Fz$ has orbit size 39. For the last code the seven orbits are those of

$$\partial'_2(1), \qquad \partial'_2(1 + x^a y^b),$$

where (a, b) ranges over

$$\{(2, 1), (8, 1), (11, 1), (13, 0), (17, 1), (25, 0)\}.$$

Each has size 156. These classifications are *outputs* of the searches, never assumptions used in pruning. The unrestricted exclusions, cycle parity, and attaining words prove Theorem 1.1. Both implementations also search the X sector directly, in addition to checking the isometry (3).

2 Scope, projection, and transfer

Definition 2.1. A strict-trinomial binary abelian double cover consists of a surjection $\pi : \widetilde{G} \to G$ of finite abelian groups with kernel of order two, and polynomials $\widetilde{A}, \widetilde{B} \in \mathbb{F}_2[\widetilde{G}]$, each with three distinct supported group elements, whose images A, B also each have three distinct supported elements. Terms are matched individually. We assume the base code has $k > 0$, and write its parameters as $[[n, k, d]]$, where $n = 2|G|$, and those of the cover as $[[2n, k_2, d_2]]$.

This class includes the rectangular BB covers of Section 1. The proofs below work in the larger abelian class. They do not apply without modification to arbitrary Tanner graph covers, to polynomials with canceled terms after projection, or to an unrestricted number of check terms.

Multiplying either check by a monomial induces a physical coordinate permutation on the complex. We may therefore normalize, compatibly in the two codes,

$$A = 1 + a_1 + a_2, \qquad B = 1 + b_1 + b_2. \tag{13}$$

The normalized support subgroup $K = \langle a_1, a_2, b_1, b_2 \rangle$ decomposes the code into isomorphic coset components. The distance is the distance of one component; positive dimension is equivalent to positive dimension of that component. The upstairs support subgroup maps onto K with kernel of order one or two. Thus a disconnected cover creates no exceptional case: a degree-one component is isomorphic to a base component, and a degree-two component obeys the arguments below.

Proposition 2.2 (Homological rank constraints)**.** *Every cover in the stated class satisfies*

$$k \le k_2 \le 2k, \qquad \operatorname{rank}(\widehat{\tau}_1) = \operatorname{rank}(\widehat{p}_1) = \frac{k_2}{2}, \tag{14}$$

where p is projection and τ is the two-sheet transfer. In particular,

$$k_2 = 2k \quad \Longrightarrow \quad d_2 \le 2d. \tag{15}$$

Proof. Let t generate the deck kernel and put $\varepsilon = 1 + t$, so $\varepsilon^2 = 0$. Projection and transfer give an exact sequence of chain complexes

$$0 \longrightarrow C \xrightarrow{\tau} \widetilde{C} \xrightarrow{p} C \longrightarrow 0.$$

For a basis element $g \in G$, the transfer is the sum of its two preimages. It is independent of the chosen preimage and has weight two.

Let $M = R/(A, B)$ and $\widetilde{M} = \widetilde{R}/(\widetilde{A}, \widetilde{B})$. Put $h = \dim M = k/2$ and $\widetilde{h} = \dim \widetilde{M} = k_2/2$. Since $\widetilde{M}/\varepsilon\widetilde{M} = M$ and $\varepsilon\widetilde{M}$ is a quotient of M,

$$h \le \widetilde{h} \le 2h, \qquad \operatorname{rank} \widehat{\tau}_0 = \dim \varepsilon\widetilde{M} = \widetilde{h} - h.$$

Exactness of the long homology sequence gives the connecting map rank

$$\operatorname{rank} \beta_1 = h - (\widetilde{h} - h) = 2h - \widetilde{h}.$$

Therefore $\operatorname{rank} \widehat{p}_1 = k - \operatorname{rank} \beta_1 = \widetilde{h}$, and then $\operatorname{rank} \widehat{\tau}_1 = k_2 - \widetilde{h} = \widetilde{h}$. When $k_2 = 2k$, transfer is injective on $H_1(C)$. Transferring a minimum base logical produces a nontrivial cover logical of weight $2d$. □

The rank formulas explain why no contradiction arises between the examples and (15). The distance-increasing example has $k_2 = k$, so some nonzero base classes are killed by transfer. The distance-decreasing example has $k_2 = 2k$, so transfer is injective, but projection still has a nonzero kernel.

Proposition 2.3 (A refined transfer bound)**.** *Let $\lambda_j(Q)$ be the least integer r such that classes represented by cycles of weight at most r span a subspace of $H_1(Q)$ of dimension at least j. Then*

$$d_2 \le 2\lambda_{k-k_2/2+1}(Q).$$

Proof. The kernel of transfer has dimension $k - k_2/2$. A spanning space of larger dimension cannot lie in that kernel, so one of its weight-bounded generating classes survives transfer. □

There is also an exact, but code-dependent, separation of the two cover sectors. Let d_{vis} be the least weight of a cover cycle whose projected class is nonzero, and $d_{\ker}$ the least weight of a nonzero cover class in $\ker \widehat{p}_1$. Then

$$d_2 = \min\{d_{\mathrm{vis}}, d_{\ker}\}, \qquad d_{\mathrm{vis}} \geq d.$$

This identity does not turn d into a universal lower bound: the second sector remains essential. Likewise a shortest base logical need not survive transfer, and subtracting a lifted boundary has no automatic Hamming-norm control. These are metric issues, not failures of the chain maps.

3 A two-sided polynomial replacement

The two distance inequalities fail, but neither distance can be held fixed while the other tends to infinity.

Theorem 3.1 (Polynomial distance comparison)**.** *For every strict-trinomial binary abelian double cover with $k > 0$,*

$$\boxed{d \leq 4d_2^3, \qquad d_2 \leq 8d^3.} \tag{16}$$

In particular, with n the number of base qubits,

$$2\left\lceil \frac{(d/4)^{1/3}}{2} \right\rceil \;\leq\; d_2 \;\leq\; 2\left\lfloor \frac{\min\{8d^3, n\}}{2} \right\rfloor. \tag{17}$$

The lower inequality $d \leq 4d_2^3$ also holds for term-preserving finite abelian quotients whose kernel is any finite 2-group.

The archived amplification argument gave $d_2 \leq 32d^3$. The constant eight in (16) is a consolidation corollary: the cofactor argument used for descent can be transported upwards through an index-at-most-two relation sublattice. It avoids the external lattice-code theorem previously used for the finite-presentation case. We give the complete argument, including the higher-rank torsion case, rather than treating this improvement as a numerical extrapolation.

3.1 Two elementary distance bounds

Lemma 3.2 (A finite check subgroup)**.** *Suppose one normalized check is supported in a finite subgroup $K \leq G$. If the code has positive dimension, then $d \leq |K|$. In particular $d \leq |G| = n/2$.*

Proof. Assume the check is A. The group algebra is free over $\mathbb{F}_2[K]$, so $\mathrm{Ann}(A)$ is generated, as an R-module, by elements supported in K, each of weight at most $|K|$. The common annihilator of A, B is nonzero: the nondegenerate identity-coefficient pairing identifies it with the dual of $R/(A, B)$. Thus multiplication by B is singular, hence nonsurjective, on $\mathrm{Ann}(A)$. Its image is an R-submodule. At least one of the indicated generators u is outside $B\,\mathrm{Ann}(A)$. The cycle $(u, 0)$ is then a nonboundary: a boundary representation would require $Bh = u$ and $Ah = 0$. □

Lemma 3.3 (Strip cleaning and cofactors)**.** *Suppose the normalized check translations factor through $T \oplus \mathbb{Z}$, where T is finite. Let R_0 be the larger of the two check-support spans in a primitive free coordinate. Every positive-dimensional finite image satisfies*

$$d \leq 4|T|R_0. \tag{18}$$

If $r_1, r_2, r_3 \in \mathbb{Z}^4$ are independent relations and $c \in \mathbb{Z}^4$ is their signed cofactor vector, every positive-dimensional finite quotient of $\mathbb{Z}^4/\langle r_1, r_2, r_3\rangle$ satisfies

$$d \le 4 \max\{\mathrm{span}(0, c_1, c_2), \mathrm{span}(0, c_3, c_4)\}. \tag{19}$$

Here span *of finitely many real numbers is their maximum minus their minimum.*

Proof. Quotient the finite image by the image of T. The quotient is cyclic, with at most $2|T|$ qubits per cell. A seam twist merely permutes the internal coordinates. Shift left and right block origins independently so that both stabilizer types have span at most R_0: if the two check supports start at $a_{\min}, b_{\min}$, shift the left block by $-b_{\min}$ and the right block by $-a_{\min}$.

If the number P of cells is less than $2R_0$, Lemma 3.2 already suffices. Otherwise partition the circle into an even number of intervals, each of length between R_0 and $2R_0$. One may start with $2\lfloor P/(2R_0)\rfloor$ intervals of length R_0 and distribute the remaining cells. If $d > 4|T|R_0$, every interval is correctable. No stabilizer generator meets two intervals of the same parity. A commuting Pauli supported on the even union therefore has separately commuting restrictions to its intervals; each restriction is a stabilizer. The even union is correctable, and so is the complementary odd union. The cleaning lemma rules out two complementary correctable regions for $k > 0$.

For completeness, cleaning is finite symplectic linear algebra: if a logical restriction to a correctable region cannot be canceled by a stabilizer restriction, a separating linear functional gives a Pauli supported in that region which commutes with the stabilizers but anticommutes with that logical, contradicting correctability. No geometric limit is involved.

For three independent rows, let τ be the gcd of the entries of c. Smith normal form gives $|T| = \tau$, and the primitive free coordinate is c/τ. Substitution into (18) gives (19). □

3.2 Presentations extracted from a minimum logical

Choose a minimum nonboundary cycle of weight w in either source code. It has circuit support. Pair the syndrome incidences at each check; there are $3w$ incidences. The resulting cubic graph on the selected qubits is connected, since a component would otherwise be a proper subcycle. It has no loops because each column has three distinct incidences. Parallel edges are allowed.

Label the three ports at a left qubit by $0, e_1, e_2$, and those at a right qubit by $0, e_3, e_4$. An edge pairing ports at qubits u, v imposes

$$p_u + c_{u,i} = p_v + c_{v,j}.$$

A spanning tree determines all relative qubit positions. The remaining edges supply relation rows in $\mathbb{Z}^4$. Let L be their integral span and put

$$H = \mathbb{Z}^4/L. \tag{20}$$

The actual support group is a quotient of H, and the selected circuit lifts to a finite-support cycle over $\mathbb{F}_2[H]$. Its qubits remain distinct, since a collision in H would be a collision in the source code. If this universal cycle were a boundary, its finite image would be a boundary; hence it is nontrivial universally as well.

Every fundamental graph cycle has length at most w. Its relation row is a sum of at most w vectors from

$$\mathcal{D} = \{\pm e_1, \pm e_2, \pm(e_1 - e_2), \pm e_3, \pm e_4, \pm(e_3 - e_4)\}. \tag{21}$$

In particular every row has Euclidean norm at most $\sqrt{2}w$. Every determinant of four vectors from $\mathcal{D}$ is 0 or ± 1: a nonzero determinant uses two vectors in each coordinate block, where the planar

determinants have those values. If r_1, r_2, r_3 are independent fundamental rows with cofactor vector c, multilinearity therefore gives

$$\max_{\delta \in \mathcal{D}} |c \cdot \delta| \le w^3. \tag{22}$$

The left side is precisely the maximum span in (19).

3.3 The index-two transport lemma

Lemma 3.4 (Transporting three relations through a double cover)**.** *Suppose three independent rows r_1, r_2, r_3 vanish in a base group G. Under the corresponding lifted translations in $\widetilde{G}$, there is a rank-three relation sublattice with cofactor vector $c' = jc$, where $j \in \{1, 2\}$ and c is the original cofactor vector, up to a common orientation sign. Consequently, if the rows obey* (22)*,*

$$d_2 \le 8w^3. \tag{23}$$

Proof. Let $K_3 = \langle r_1, r_2, r_3 \rangle \cong \mathbb{Z}^3$. Each row maps under lifted translations to the deck kernel $D \cong \mathbb{Z}_2$. This defines a homomorphism $\nu : K_3 \to D$. Set $K_3' = \ker \nu$. Its index $j = [K_3 : K_3']$ is one or two, and all its elements are actual cover relations.

For an integral basis change from K_3 to K_3', the determinant is $\pm j$. Taking the third exterior power multiplies the cofactor vector by the same scalar, hence $c' = \pm jc$. The cover is a finite quotient of $\mathbb{Z}^4/K_3'$. Lemma 3.3 and (22) yield $d_2 \le 4jw^3 \le 8w^3$.

Explicitly, if the parity vector is nonzero and r_i has odd voltage, a basis of the kernel is $2r_i$ together with $r_j - \nu(r_j) r_i$ for $j \ne i$. Its index is two, not eight. There is no need to double all three rows separately, nor to lift the original logical as a cycle. □

This lemma treats both finite and virtually cyclic incidence presentations: even if L has rank four, we use only three independent relations. Conversely, for descent from a cover, the same three rows already vanish in the base, so (19) gives $d \le 4w^3$ without an index increase.

3.4 Pure circuits and higher-free-rank structure

For a circuit contained in one block, use only the two translations of that check. A strict trinomial is a nonzerodivisor in its support-generated group algebra whenever that group has positive free rank. Indeed, choose a free-coordinate direction with a unique extremal term. When two free positions coincide, choose the remaining singleton as the extremum; its coefficient is a torsion monomial, hence a unit. Comparing extremal degrees proves injectivity of multiplication.

A pure universal cycle therefore forces its two-generator presentation to be finite. Two independent fundamental rows and Hadamard's inequality give group order at most $2w^2$. The image of this subgroup has size at most $2w^2$ downstairs and its preimage at most $4w^2$ upstairs. Lemma 3.2 gives the corresponding target distance bounds. From now on the source circuit meets both blocks.

Write $H = T \oplus \mathbb{Z}^r$. Split the odd torsion into characters after extending scalars to an algebraic closure of $\mathbb{F}_2$, and filter the 2-primary group algebra by its nilpotent augmentation ideal. Each associated graded layer is a free module over the Laurent polynomial ring of the free part.

Lemma 3.5 (Higher-rank classification)**.** *If neither check has a point as its free support, nonzero universal first homology is impossible for $r \ge 3$. For $r = 2$, it forces, after monomial gauges, term permutations, and possibly exchanging the checks,*

$$H = T \oplus \langle x, y \rangle, \qquad A = 1 + x + y, \quad B = 1 + tx^q + uy^q, \quad q = 2^s, \quad T = \langle t, u \rangle. \tag{24}$$

Only the trivial character of T_{odd} can support this homology. Moreover, for a weight-w incidence presentation,

$$|T| \le \frac{2w^2}{1+q^2}. \tag{25}$$

Proof. In an odd-character component, reduce modulo the 2-primary augmentation ideal. If the two reduced Laurent polynomials are nonzero and coprime, every cycle in the Laurent domain is a boundary. Lift its boundary preimage and subtract it; repeat in the successive augmentation layers. Nilpotence terminates the procedure. Thus coprime reductions imply exactness also before reduction.

A non-point trinomial support cannot reduce to zero: if two free positions coincide, the singleton coefficient remains nonzero. The Newton polytope of a product is the Minkowski sum of the factor polytopes. A nontrivial segment cannot be a summand of a nondegenerate triangle. Every non-point summand of a triangle is a positively homothetic triangle: its edge directions are among the three triangle directions, and the positive closing relation fixes their relative lengths. Therefore a common factor of triangular trinomials forces positively homothetic coplanar support triangles. Nonparallel segments, or a segment and a triangle, have no common nonunit factor. If one support is a point, it is torsion-supported and the finite-check lemma applies instead. Otherwise combined free rank at least three gives coprime reductions in every component, an impossibility for the chosen cycle.

For $r = 2$ with neither free support a point, the nonregular case must be two homothetic triangles. Their exponents can be written

$$(0, pv_1, pv_2), \qquad (0, qv_1, qv_2), \qquad \gcd(p, q) = 1.$$

Bezout's identity shows v_1, v_2 are integral and generate the free lattice. They are therefore a basis. In these coordinates the reduced polynomials are scalar Fermat polynomials. Write $p = 2^a p_0$, with p_0 odd. In characteristic two, the polynomial is a 2^a-th power of a scalar Fermat polynomial of odd degree p_0. Its projective homogenization is smooth: the three partial derivatives cannot all vanish. A reducible plane projective curve has intersecting components and hence a singular point, so this odd-degree polynomial is absolutely irreducible. The degree-one case is immediate. The same argument applies to q. A common factor forces equal odd parts; coprimality of p, q then forces both odd parts to equal one, and one of p, q is one. This proves the form (24).

The lifts x, y of the smaller triangle split the free part. The remaining generators are the torsion elements t, u. Since the reduced A is $1 + x + y$, divisibility of the reduced B by A is equivalent to $\chi(t) = \chi(u) = 1$. They generate the torsion, so the odd character is trivial.

Finally all relation rows in these normalized port coordinates have form $(-q\beta_1, -q\beta_2, \beta_1, \beta_2)$. Projection to $\beta \in \mathbb{Z}^2$ identifies the saturated relation lattice with $\mathbb{Z}^2$, and $|T|$ is the index of the actual relation lattice there. Two independent fundamental rows have β norms at most $\sqrt{2}w/\sqrt{1+q^2}$. Their determinant bounds this index from above, giving (25). Changing reference ports does not invalidate the norm estimate: each traversed vertex still contributes a difference of two normalized ports from (21). □

3.5 A short nonboundary in an active torsion–Frobenius summand

Lemma 3.6 (Nilpotent-twist witness)**.** *Let a finite abelian code have checks $A = 1 + x + y$, $B = 1 + tx^{2^s} + uy^{2^s}$. Put $T_0 = \langle t, u \rangle$, let O be its odd subgroup, and let $E = \sum_{o \in O} o$. If $ER/(A, B) \neq 0$, then*

$$d \le |T_0|(1 + 3^s). \tag{26}$$

The active-summand assumption is essential; global $k > 0$ alone is not a replacement for it.

Proof. Work in the E summand. Its 2-primary torsion is

$$K = \langle g \rangle \times \langle h \rangle \cong C_M \times C_N, \qquad N \mid M,$$

where M, N are powers of two. Put $e = 1 + g$, $f = 1 + h$, and $q = 2^s$. For $M > 1$, write the 2-parts of t, u as $g^a h^b, g^c h^d$. At least one of a, c is odd because these two elements generate K. Put

$$C = B + A^q = eF + fH_0.$$

Modulo (e, f), the residue of F is one of x^q, y^q, or $x^q + y^q$. At every common maximal ideal of A, B in the active summand it is a unit: the first two are monomials and the last is one when $A = 0$.

Define a globally supported polynomial

$$S = Ee^{M-N}C^{N-1}, \qquad W = (SA^{q-1}, S). \tag{27}$$

Since $EC^N = Ee^N F^N$, we have $CS = 0$ and hence $\partial_1 W = 0$. For $M = N = 1$, take $S = E$; then $EC = 0$ directly.

To prove nontriviality, localize at a common maximal ideal in the active summand. The associated graded ring for $I = (e, f)$ is

$$(R/I)[e, f]/(e^M, f^N).$$

This follows from a free group-coset basis over $\mathbb{F}_2[K]$; localization and an idempotent summand preserve that freeness. The initial form of C is $e\overline{F} + f\overline{H}_0$, with $\overline{F}$ a unit. Substitution of this expression for e, leaving f fixed, is an automorphism of the graded ring. Both directions preserve the truncation relations because $N \mid M$. Thus its annihilator is generated by its $(M-1)$-st power. Lifting through the finite I filtration gives

$$\mathrm{Ann}(C) = C^{M-1}R, \qquad \mathrm{Ann}(C^{M-1}) = CR.$$

Explicitly, subtract a multiple of C^{M-1} that matches the lowest nonzero graded part of an element killed by C, then iterate at higher filtration. The same argument proves the second identity. Also

$$C^{M-1} = F^{M-N}e^{M-N}C^{N-1},$$

so locally S is a unit multiple of C^{M-1}. Hence $\mathrm{Ann}(C) = SR$ and $\mathrm{Ann}(S) = CR$. The trivial-K case has the same conclusions with $C = 0, S = 1$.

If $W = (Br, Ar)$ were a boundary, then $Ar = S$ and $Br = SA^{q-1}$, whence $Cr = 0$. Write $r = Sv$. The equation $Ar = S$ now implies $S(Av + 1) = 0$, so $Av + 1 \in CR$. This is impossible: A, C belong to the chosen maximal ideal but $Av + 1$ is a unit.

Finally e^{M-N} has M/N terms, C has at most four terms, and N is a power of two. Frobenius gives $\mathrm{wt}(C^{N-1}) \le N^2$ and $\mathrm{wt}(A^{q-1}) \le 3^s$. Thus $\mathrm{wt}(S) \le |O|MN = |T_0|$ and $\mathrm{wt}(W) \le |T_0|(1 + 3^s)$. □

3.6 Why the active summand survives in the required direction

The group-algebra map of a finite abelian 2-group quotient has nilpotent kernel: it is generated by commuting nilpotents $1 + t$ for kernel elements t. It therefore induces a bijection of maximal ideals. A common maximal ideal of the two checks exists in a specified odd idempotent summand downstairs exactly when one exists in the corresponding summand upstairs.

For a source circuit with presentation (24), Lemma 3.5 says that every nontrivial universal homology component lies in the trivial odd character. For E the odd norm, the other components of the chosen universal cycle are boundaries, so its finite source class equals its E component and is

nonzero. Thus the source E summand encodes. The preceding nilpotent-kernel observation gives the active target summand. This is not an assumption that the particular logical projects or transfers nontrivially.

For descent, the target torsion image has order at most $|T|$. For ascent, one may construct the universal lifted presentation directly. Let $f : \mathbb{Z}^4 \to G$ and $\widetilde{f} : \mathbb{Z}^4 \to \widetilde{G}$ record the normalized translations, and let L be the source relation lattice. Then

$$L' = L \cap \ker \widetilde{f}, \qquad [L : L'] \le 2, \qquad \widetilde{H} = \mathbb{Z}^4/L' \longrightarrow H$$

has finite kernel of order at most two. It also maps onto the actual cover support subgroup. Its torsion has order at most $2|T|$. Choosing lifts of the free basis gives the same form $A' = 1 + x' + y'$, $B' = 1 + t'(x')^q + u'(y')^q$. The odd torsion is canonically unchanged by the 2-kernel, so the active-summand argument applies.

Combining Lemma 3.6 with (25), the distance in the target is at most

$$j|T|(1 + 3^s) \le 2jw^2 \frac{1 + 3^s}{1 + 4^s} \le 2jw^2, \tag{28}$$

where $j = 1$ for descent and $j \le 2$ for ascent. If a check is torsion-supported instead, two independent relation rows bound $|T| \le 2w^2$ directly by minors and Hadamard, and Lemma 3.2 gives the same target bound. A torsion-supported check cannot occur with combined free rank above two, since the other check supplies at most two free directions.

3.7 Completion of the proof

Proof of Theorem 3.1. For descent take $w = d_2$. Pure circuits give $d \le 2w^2$. A mixed presentation with relation rank at least three gives $d \le 4w^3$ by (19)–(22). Relation rank at most one would give free rank at least three and hence no universal nonboundary, by Lemma 3.5. In the remaining rank-two case, (28) or the finite-check argument gives $d \le 2w^2$. This proves $d \le 4d_2^3$. Only the active-summand step uses the 2-group nature of the kernel, so this conclusion holds for all finite 2-group kernels.

For ascent take $w = d$. Pure circuits give $d_2 \le 4w^2$. In relation rank at least three use Lemma 3.4, giving $d_2 \le 8w^3$. Free rank at least three is again impossible for the source nonboundary. In free rank two, (28) gives $d_2 \le 4w^2$. Since $w \ge 2$, all cases satisfy $d_2 \le 8d^3$.

Finally apply Lemma 3.2 to the cover, whose group has n elements, and use evenness of all cycle weights to round both endpoints in (17). □

4 Ambient size and truncated distortion

The polynomial bounds can be combined with a geometry-of-numbers estimate that needs no assumption about a preferred compact sector.

Theorem 4.1 (Ambient distance)**.** *Every positive-dimensional strict-trinomial binary abelian code of length n satisfies*

$$d \le \min\left\{\frac{n}{2},\ C_0 n^{3/4}\right\}, \qquad C_0 = \frac{16}{3}\left(\frac{12}{\pi^2}\right)^{3/4}. \tag{29}$$

Proof. Restrict to the normalized support group K, of order $M \le n/2$. Its full relation lattice $\Gamma = \ker(\mathbb{Z}^4 \to K)$ has determinant M. Give $\mathbb{R}^4$ the positive-definite metric

$$Q = \begin{pmatrix} 1 & 1/2 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 1/2 & 1 \end{pmatrix}, \qquad \det Q = 9/16.$$

Each port difference in $\mathcal{D}$ has Q norm one. For three independent relation rows and their cofactor vector c, Cauchy–Schwarz and the Gram/cofactor identity give

$$\max_{\delta \in \mathcal{D}} |c \cdot \delta| \le \sqrt{c^T Q^{-1} c} = \frac{1}{\sqrt{\det Q}} \sqrt{\det(r_i Q r_j^T)_{i,j=1}^3} \le \frac{4}{3} \prod_{i=1}^{3} \|r_i\|_Q.$$

Lemma 3.3 therefore bounds distance by $\frac{16}{3} \prod_{i=1}^3 \|r_i\|_Q$.

The Q unit ball has volume $2\pi^2/3$. For the successive minima $\lambda_1 \le \cdots \le \lambda_4$ of Γ, Minkowski's second theorem [2, Chapter VIII] gives

$$\prod_{i=1}^{4} \lambda_i \le \frac{24M}{\pi^2}, \qquad \lambda_1\lambda_2\lambda_3 \le \left(\prod_{i=1}^{4} \lambda_i \right)^{3/4}.$$

Choose three independent vectors attaining the first three successive minima and substitute. This gives $d \le \frac{16}{3}(24M/\pi^2)^{3/4} \le C_0 n^{3/4}$. Lemma 3.2 supplies the other endpoint. □

Define $\mathcal{C}_N$ to be the set of the stipulated rectangular BB double covers with positive base dimension and base length $n \le N$, and put

$$\gamma_-(N) = \inf_{\mathcal{C}_N} \frac{d_2}{d}, \qquad \Gamma(N) = \sup_{\mathcal{C}_N} \frac{d_2}{d}.$$

The same estimates hold when the collection is enlarged to all finite abelian covers. Bounds for an empty collection are not asserted.

Corollary 4.2 (Explicit truncated bounds)**.** *For a nonempty collection,*

$$\boxed{\max\{N^{-2/3},\ c_- N^{-1/2}\} \le \gamma_-(N), \qquad \Gamma(N) \le c_+ \sqrt{N},} \tag{30}$$

where

$$c_- = \frac{\pi 3^{1/6}}{2^{13/3}} \simeq 0.1871573, \qquad c_+ = 2^{3/2} C_0^{2/3} \simeq 9.5203204.$$

The counterexamples also give

$$\Gamma(N) \ge \frac{7}{3} \quad (N \ge 78), \qquad \gamma_-(N) \le \frac{6}{7} \quad (N \ge 156).$$

Proof. From $d \le 4d_2^3$,

$$\frac{d_2}{d} \ge 4^{-1/3} d^{-2/3}.$$

Use $d \le n/2$ for the $n^{-2/3}$ bound, or $d \le C_0 n^{3/4}$ for the $c_- n^{-1/2}$ bound. For amplification, apply the ambient estimate to the length-$2n$ cover and put $K_0 = 2^{3/4} C_0$. Then

$$\frac{d_2}{d} \le \min\{8d^2,\ K_0 n^{3/4}/d\} \le (8d^2)^{1/3} (K_0 n^{3/4}/d)^{2/3} = c_+ \sqrt{n}.$$

Take the infimum or supremum with $n \le N$. The opposite inequalities follow from the explicit base lengths of Theorem 1.1. □

The exact expression for c_+, not its printed decimal, defines the constant. The size threshold $N \geq 156$ for the degradation example follows from the now-available finite certificate; it does not require an inherited infinite-family threshold.

Combining the preceding theorems gives a fully explicit size-aware answer in the notation $d_{\text{lower}} \leq d_2 \leq d_{\text{upper}}$:

$$d_{\text{lower}}(d) = 2\left\lceil \frac{(d/4)^{1/3}}{2} \right\rceil, \tag{31}$$

$$d_{\text{upper}}(d,n) = 2\left\lfloor \frac{\min\{8d^3, n, C_0(2n)^{3/4}\}}{2} \right\rfloor. \tag{32}$$

When $k_2 = 2k$, also take the minimum with $2d$ in (32). None of these formulas claims an optimal constant or exponent. In particular (30) is not a two-sided asymptotic equivalence for either extremal function, and it does not prove bounded $\sup d_2/d$ or a positive $\inf d_2/d$ independent of size.

5 Companion results and boundaries of the present conclusions

The repository preserves two longer research branches in addition to the counterexample certificates. They contain further symbolic arguments, finite classifications, exact optimization certificates, and failed approaches. We distinguish their roles here. The main proofs in Sections 2–4 do not depend on a small-weight classification, an unverified decoder output, or the stronger compact-metric results summarized below.

The retained incidence classifications also give the useful specialized bounds

$$d = 6 \implies d_2 \leq 112, \qquad d_2 = 8 \implies d \leq 96$$

[6, 7]. Their full arguments and classification code are included as companion notes. The fresh audit rechecks retained algebraic certificates, but does not rerun both expensive complete weight-six classification generators. Earlier weight-two/four assertions and the original weight-six descent note have separate provenance or starting assumptions, recorded in the repository claim map; they are not hidden dependencies of the general theorem.

5.1 An exact higher-rank compact metric

For the support-generated normal form

$$T \oplus \mathbb{Z}^2, \qquad A = 1 + X + Y, \qquad B = 1 + hX^{2^s} + kY^{2^s}, \qquad T = \langle h, k \rangle,$$

the companion compact-metric note [3] gives the exact minimum weight of a nonzero universal compact class:

$$\delta_{\text{comp}} = o\, 2^{a-b} 3^b (3^s + 1), \tag{33}$$

where $o = |T_{\text{odd}}|$ and $T_2 \cong C_{2^a} \times C_{2^b}$, $a \geq b \geq 0$. It constructs a minimum representative generating the compact module, and gives compatible double-cover replacement ratios $1, 3/2, 2$ in the stated normal-form setting. Its proof uses a squaring-and-coset lower bound for a finite-group kernel, two generic evaluations to bound the two physical blocks separately, and an attaining module generator. The retained audit checks 2,322 explicit primal/dual replacement certificates.

This is a factor-two theorem in its specified higher-rank regime, not a global factor-two theorem for arbitrary BB double covers. Specialization may create homology not represented by compact

cycles, and a nonzero universal class can become a finite boundary. To apply a compact result to code distance, finite survival or a winding exclusion must be proved. The polynomial comparison above avoids replacing finite homology by compact homology.

5.2 A rank-one relative Laurent optimization

The other branch reduces a remaining family to a physical, rather than abstract, polynomial minimization. Let

$$S = \mathbb{F}_2[z^{\pm 1}], \quad c = 1 + z + z^2, \quad B = 1 + z^E + z^O, \quad E = 6r + 2, \quad O = 6t + 1,$$

with three distinct terms, and set $b = B/c$, $P = (B + c)/c^2 = (b + 1)/c$. For the stated finite-survival criterion, the relative Laurent minima are

$$\mu_0 = \min_{c \nmid F} \big(\mathrm{wt}(bF) + \mathrm{wt}(F)\big), \tag{34}$$

$$\mu_+ = \min_{\substack{F,H \in S \\ c \nmid F}} \left[\mathrm{wt}((b+P)F + bH) + \mathrm{wt}(PF + bH) + \mathrm{wt}(F + H) + \mathrm{wt}(H)\right]. \tag{35}$$

All Laurent messages F and all boundary adjustments H remain unrestricted. The condition $c \nmid F$ is part of the relative problem; it must not silently be replaced by nonzero universal homology alone.

The physical source of the formula is particularly simple. In $\mathcal{R} = S[s]/(s^2 + 1)$, set $e = 1 + s$, $\Theta = c + e$, and $D = b + eP$. The lifted checks are

$$\widetilde{A} = \Theta = s + z + z^2, \qquad \widetilde{B} = \Theta D = s + z^E + z^O.$$

Since $\Theta^2 = c^2$ is a nonzerodivisor, all universal cycles are (DV, V) and universal boundaries have $V \in \Theta\mathcal{R}$. Writing $V = F + eH$ and expanding in the physical sheet basis $1, s$ produces exactly the four weights in (35). Thus the metric is explicit rather than assigned after a non-isometric normal-form change.

The supplied full proof [4] establishes

$$\mathrm{span}(B) \le 4\,\mathrm{wt}(F)\,\mathrm{wt}(bF), \qquad \mu_+ \le \mu_0^2 + 2, \tag{36}$$

with the $+2$ absent when $E, O > 0$. It also shows that the twelve unit residues of $S/(c^2)$ are physical monomial translates, yielding an exact single-coset minimization with an unrestricted boundary variable. These statements concern the stated rank-one subclass and require the accompanying stabilization hypotheses before becoming equalities of finite code distances. They are not substitutes for a global linear amplification theorem or a quadratic degradation theorem.

5.3 Exact optimization records and negative controls

The companion rank-one package contains all-degree Bellman certificates for relative minima 8 and 14 for $A = 1 + z^2 + z^7$, $B = 1 + z + z^8$ with the specified constant-term twists. Its stabilization theorem gives $[[2L, 4, 8]] \to [[4L, 4, 14]]$ for odd $L \ge 105$ divisible by three [5]. This is useful independent evidence about complete optimizations; its ratio is not an improvement on 7/3.

Several large values in the research history are deliberately *not* promoted to distance records. A sector-restricted minimum 16 coexists with full admissible minimum 14. A fixed-message minimum $74N$ lies in a universal class admitting a weight-fourteen representative. A relaxed family with unbounded amplification uses a five-term lifted check and violates the strict three-term matching condition. Such records refute shortcuts involving a preferred message or sector; they do not refute an inequality for the full target class. The repository's claim map links each assertion to its proof and exact audit scope.

5.4 Open quantitative targets

For the stated class, the finiteness of the unrestricted amplification constant $C_* = \sup d_2/d$ and positivity of the unrestricted degradation ratio $\inf d_2/d$ are not settled by the supplied results. A global quadratic improvement $d \le Cd_2^2$ is also not established. In the fixed-c rank-one problem, $\mu_+ \le 3\mu_0$ remains unproved and unrefuted in this record. Any proposed counterexample must bound the complete minimum over F, H, not a selected class or fixed-message slice.

The cubic inequalities impose concrete restrictions on further constructions. An unbounded amplification sequence must have $d \to \infty$ along a subsequence. A vanishing degradation sequence must have $d_2 \to \infty$ as $d \to \infty$. These conclusions do not decide either universal constant question.

A Why the finite distance certificates are exhaustive

This appendix specifies the mathematical decision problem independently of the stored outputs. All four standalone programs regenerate their input matrices from the displayed periods and check terms. No program reads an asserted distance, a supplied list of circuits, another verifier, or a heuristic decoder output as evidence of exclusion.

A.1 Reduction to circuits and two translation roots

For a binary matrix H, a circuit is a nonempty inclusion-minimal support of a vector in $\ker H$. Every binary cycle is a disjoint union of circuits: take a nonempty minimal subcycle, remove it, and iterate. If a nonboundary cycle has minimum weight, it must itself be a circuit. Otherwise all smaller constituent circuits would be boundaries, making their sum a boundary.

Each BB column has weight three, so the parity of the syndrome weight equals the parity of the selected support size. Every cycle has even weight. It is therefore enough to exclude nonboundary circuits through weight 4, 12, and 10 for the three codes respectively. Translation preserves the checks and boundaries. A circuit meeting the left block has a translate containing the left identity qubit; one meeting the right block has a translate containing the right identity qubit. The union of all translates of the circuits found at these two roots is the full circuit set up to the cutoff. Redundant roots and translations are deduplicated as sorted physical-coordinate tuples.

A.2 First implementation: first-present branching

A state records chosen variables U, excluded variables E, and syndrome $s = \sum_{j \in U} H_j$. The root starts in U. At cutoff b, write $r = b - |U|$ for the remaining budget. If $s \neq 0$, the following are necessary conditions for completion: $r > 0$, $|s| \le 3r$, and every odd check has an available incident variable. Violation of any condition prunes the branch.

Choose an odd check and order its undecided incident variables $v_1, \dots, v_t$. A completing circuit must select at least one. Branch i selects v_i and excludes $v_1, \dots, v_{i-1}$, leaving later variables undecided. In a prescribed target circuit, its first present variable determines exactly one branch. Thus the branching does not omit a circuit.

At zero syndrome, test whether the selected cycle is a boundary and stop that branch. This cannot truncate a larger circuit containing U, because a circuit has no proper nonempty zero-syndrome subset. The leaf is recorded as a circuit exactly when the selected columns have rank $|U| - 1$. Indeed, their sum is zero; nullity one is equivalent to the absence of a proper nonempty subcycle. Boundary membership is tested by exact binary elimination against the image of the boundary matrix, with a preimage recovered when needed.

A.3 Second implementation: all odd local assignments

The second program uses polynomial supports represented by finite sets and symmetric difference, not the first program's integer-bitset routines. At a selected odd check, let T be the undecided incident variables. Enumerate *every odd-cardinality subset* $V \subseteq T$ whose size fits the remaining budget, select V, and exclude $T \setminus V$. This fixes all of T at once. Every completing circuit has a unique intersection V with T, and that intersection is odd. The same necessary pruning conditions and zero-syndrome stopping argument apply. The search trees and node counts differ from the first implementation.

For a cyclic code with syndrome map (P, Q) over $\mathbb{F}_2[x]/F_L$, put

$$g = \gcd(P, Q, F_L), \quad a = P/g, \quad b = Q/g, \quad f = F_L/g.$$

Compute a polynomial Bezout identity $Ua+Vb+Wf = 1$. A candidate (u, v) cannot be a boundary unless both coordinates are divisible by g. Otherwise set

$$h = U(v/g) + V(u/g) \pmod{f}.$$

It is a boundary exactly when $Qh = u$ and $Ph = v$ modulo F_L. Sufficiency gives an explicit preimage. For necessity, if $(u, v) = (Qh_0, Ph_0)$, division by g and the Bezout identity give $h = h_0 \pmod{f}$, which yields the same boundary. This works without square-free assumptions. For the two-variable code, write $P = P_0 + yP_1$, $Q = Q_0 + yQ_1$, with $Q_1 = x^2$ a unit. For a boundary preimage $h = h_0 + yh_1$, eliminate

$$h_1 = Q_1^{-1}(u_0 + Q_0 h_0).$$

The remaining equations for h_0 are

$$(Q_0^2 + Q_1^2)h_0 = Q_1 u_1 + Q_0 u_0, \qquad (P_0 + P_1 Q_1^{-1} Q_0)h_0 = v_0 + P_1 Q_1^{-1} u_0.$$

The same polynomial congruence solver decides these equations, followed by a direct check of both complete boundary coordinates. Necessity follows from elimination; sufficiency follows from the reconstructed preimage. Thus this implementation does not reuse the bitset boundary-span test. The dual-number witness separately proves nontriviality without either search.

A.4 Boundary orbits and checks in both sectors

For the period-78 code, the 741 short circuits split into ten translation orbits. Nine have size 78, and one has size 39. Representatives are the boundaries of

$$1; \quad 1 + x^a \ (a = 2, 8, 11, 13, 17, 25); \quad A; \quad B; \quad H,$$

where H is the polynomial in Section 1. Their respective circuit-weight totals are 78 at weight six, 468 at weight ten, and 195 at weight twelve. For the $(78, 2)$ code, the $1,092$ circuits through weight ten form seven size-156 orbits: the boundary of 1 and the boundaries of $1 + x^a y^b$ for

$$(a, b) = (2, 1), (8, 1), (11, 1), (13, 0), (17, 1), (25, 0).$$

The totals are 156 at weight six and 936 at weight ten. These orbit descriptions are independently reconstructed from the search results, not used as a completeness assumption.

Each program also enumerates the X sector directly, checks all commutation entries and ranks, verifies the inversion-and-block-swap matrix identities, and compares the complete X circuit set with the transformed Z set. Each code's explicit Z logical and its X image attain the stated even distance. A mixed Pauli logical cannot be lighter: at least one of its CSS components is a nontrivial pure-sector logical, and its support is contained in the union support of the mixed operator.

A.5 Canonical output and the shared middle code

Matrices are ASCII with header `GF2MATRIX v1`, followed by dimensions and then one binary row per line. The unified runner compares all independently regenerated matrix files and sorted circuit sets. It additionally checks that the period-78 cover in the first counterexample is literally the period-78 base in the second, including both matrices and all 741 short circuits. The upper package uses a two-line circuit header and the lower package does not; the comparison strips only that documented header before comparing supports. SHA-256 hashes identify exact files but are not substitutes for the exhaustion proof above.

B Reproduction and evidence boundaries

The accompanying repository separates new mathematical exposition, unchanged certificate implementations, archived exploratory work, and freshly generated execution reports. The principal commands, run from the repository root, are

```
python -m unittest discover -s tests -v
python scripts/reproduce.py --profile core --output checks/local
python scripts/reproduce.py --profile extended --output checks/extended
python scripts/build_paper.py
```

The core profile needs only the Python standard library. It runs the new interface tests and cofactor-transport identities, executes all four standalone certificate programs, and cross-compares their matrices and full circuit outputs. An explicit timeout is recorded as incomplete, never as a successful exclusion. The original verifiers have no shared dependency on the convenience package.

The extended profile additionally requires SymPy and a C++ compiler and invokes the two archived audit drivers on materialized working copies. It reruns the retained exact verification commands, including higher-rank replacements and rank-one optimization certificates. It does not claim to regenerate every exploratory scan or every expensive classification from scratch. The precise commands, return codes, execution times, interpreter information, source hashes, and output hashes are retained. Machine checks of determinant identities support implementation review; they do not constitute a formal proof of the universal theorems.

The input inventory records each member of the three supplied archives with a content hash and either its retained destination or an explicit omission reason. Duplicate nested archives, generated caches and binaries, private conversation exports, and an unrelated third-party source PDF are excluded from the public package. The cited source paper is identified by version and hash instead of redistributed. Archived source files are not silently rewritten to make their historical status agree with the new manuscript.

Two distinctions matter for independent review. First, older research summaries referred to unavailable early certificate bundles; the later supplied certificate archive repairs the *two finite counterexamples*, but does not regenerate every earlier infinite-family or small-weight classification. Second, the constant 32 belongs to the archived general upper-bound proof, whereas the constant 8 and improved truncated amplification constant are derived in this manuscript by Lemma 3.4. These are proof-based consolidation additions, not previously recorded numerical results. The author list, affiliations, acknowledgments, software license, and publication metadata must be supplied or approved by the research authors before public submission.

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[2] J. W. S. Cassels, *An Introduction to the Geometry of Numbers*, Springer, Classics in Mathematics, 1997 reprint, Chapter VIII.

[3] Companion research note, *Linear compact replacement*, archived amplification branch, round 06 (2026). Included with this work at `research/amplification/research/06_linear_compact_replacement/REPORT.md`, with exact primal/dual certificate generators and verification code.

[4] Companion research note, *Fixed-c product bound*, archived degradation branch, round 09 (2026). Included with this work at `research/degradation/research/09-fixed-c-product-bound/proof.md`.

[5] Companion research note, *Rank-one amplification*, archived degradation branch, round 07 (2026). Included with this work at `research/degradation/research/07-rank-one-amplification/proof.md`, with all-degree Bellman certificates and a finite stabilization proof.

[6] Companion research note, *BB distance-six obstruction*, archived amplification branch, round 04 (2026). Included at `research/amplification/research/04_distance6_obstruction/REPORT.md`, with independent complete incidence classifiers.

[7] Companion research note, *A weight-eight quotient bound for strict-trinomial BB codes*, archived degradation branch, round 02 (2026). Included at `research/degradation/research/02-weight-eight/proof.md`.